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**Sources: Own elaboration**

## **1. Abstract**

This study examines socioeconomic, health, educational, and environmental development across nine countries: Switzerland, Australia, Sweden, France, Poland, China, Brazil, Kenya, and Chad over the period 2000 - 2023. Using World Bank Open Data, the analysis focuses on key indicators including life expectancy, GDP per capita, health expenditure, fertility, infant mortality, secondary and tertiary education enrollment, CO<sub>2</sub> emissions, and access to basic sanitation. The study employs a quantitative research approach with descriptive statistics, trend analysis, scatterplots, and correlation mapping to identify patterns and relationships across countries of varying development levels. Findings reveal consistent improvements in life expectancy, strong positive associations between health spending, education, sanitation, and population well-being, and mixed relationships between economic growth and environmental impact. The research highlights persistent disparities between high, middle, and low-income countries, emphasizing the importance of coordinated policy interventions. Recommendations focus on integrated investment in health, education, sanitation, and sustainable economic development to support balanced, long-term progress and align with global sustainable development goals.

## 2. Introduction

Over the past two decades, the world has experienced major shifts in economic growth, social development, and environmental sustainability. Globalization and technological change have raised living standards, yet inequalities between developed and developing countries remain significant. Understanding these differences and long-term trends is vital for shaping policies that promote inclusive progress.

This report analyzes nine countries: Switzerland, Australia, Sweden, France, Poland, China, Brazil, Kenya, and Chad during the period 2000-2023. These nations were selected to represent a range of development levels, from advanced economies to emerging and low-income countries. The analysis uses World Bank Open Data, which provides consistent annual observations across multiple indicators.

The dataset includes life expectancy at birth, GDP per capita (constant 2015 US\$), fertility rate, infant mortality, health expenditure, school enrollment (secondary and tertiary), CO<sub>2</sub> emissions per capita, and access to basic sanitation. Together, these variables capture the main aspects of economic strength, human development, and environmental performance.

The main goal of this study is to analyze and visualize changes over time, identify relationships among indicators, and compare progress between countries. Using panel data, which combines both time and cross-country dimensions, the report highlights shared global trends and distinct national paths.

Ultimately, this research contributes to understanding how social and economic progress align with sustainable development goals (SDGs). By comparing diverse countries over two decades, it provides insights into the dynamics of growth, health, education, and environmental change in the 21st century.

## 3. Literature Review

Previous research highlights the strong links between economic growth, health, education, and environmental sustainability. Bloom and Canning (2008) show that better population health supports productivity and long-term economic growth, while Barro (2013) emphasizes education as a key driver of human capital and income expansion.

Empirical studies confirm these connections in different regions. Țarcă, and Moscalu (2024) find that GDP per capita, health spending, and environmental conditions significantly affect life expectancy in Eastern Europe. Similarly, Kutlu and Örün (2022) demonstrate that carbon emissions and income levels influence health expenditure in OECD countries.

The environmental dimension is underscored by the IPCC (2023), which warns that economic growth and industrialization continue to drive CO<sub>2</sub> emissions and climate risks. Together, these studies provide the foundation for this report's analysis of economic, social, and environmental indicators across countries from 2000 to 2023.

## 4. Methodology and Data Description

This chapter outlines the research methodology employed in the study, and procedures used to collect data. It also provides a description of the data sources, variables, and descriptive statistics to ensure the reliability and validity of the findings.

### 3.1. Methodology

This study follows a quantitative research approach using secondary data obtained from the World Bank Open Data platform. After gathering the dataset, the information was carefully cleaned and prepared for analysis. This process included checking for missing values, correcting irregular entries, ensuring consistent units of measurement, and aligning time series across variables.

This study uses a balanced panel dataset consisting of 216 observations from 9 countries over the period 2000-2023. Each country is observed annually for 24 years, ensuring consistent coverage across all cross-sectional units.

Once the dataset was validated, descriptive statistics (mean, median, standard deviation, minimum, maximum, and skewness) were calculated to summarize the distribution of each variable. These statistical measures provided the foundation for further exploration through graphs, trend analysis, and correlation mapping, allowing us to examine patterns and relationships across economic, social, and environmental dimensions.

### 3.2. Data Description

The indicators used in this study were collected from the World Bank Open Data database, which provides reliable and internationally comparable statistics. These variables were selected to capture key dimensions of economic performance, social well-being, and environmental sustainability across the nine countries analyzed.

| Variable Name                                       | Definition                                                                                |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Life expectancy at birth, total (years)</b>      | Average number of years a newborn is expected to live under current mortality conditions. |
| <b>GDP per capita (constant 2015 US\$)</b>          | Total economic output per person, adjusted for inflation to 2015 US dollars.              |
| <b>Health expenditure per capita (current US\$)</b> | Total annual health spending per person, measured in current US dollars.                  |
| <b>School enrollment, secondary (% gross)</b>       | Total secondary-level enrollment as a percentage of the relevant age group.               |
| <b>School enrollment, tertiary (% gross)</b>        | Total higher-education enrollment as a percentage of the relevant age group.              |
| <b>Fertility rate, total (births per woman)</b>     | Average number of children a woman is expected to have during her lifetime.               |

|                                                                                                         |                                                                                                  |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Mortality rate, infant (per 1,000 live births)</b>                                                   | Number of infants dying before age one per 1,000 live births each year.                          |
| <b>Carbon dioxide (CO<sub>2</sub>) emissions excluding LULUCF per capita (t CO<sub>2</sub>e/capita)</b> | Annual CO <sub>2</sub> emissions per person from major sectors, excluding land use and forestry. |
| <b>People using at least basic sanitation services (% of population)</b>                                | Share of population with access to improved, non-shared sanitation facilities.                   |

Figure 1. Variables description

Together, these variables provide a comprehensive view of human development, economic progress, and environmental change over time.

### 3.3. Descriptive Statistics

To better understand the distribution and characteristics of each variable, descriptive statistics were computed. These metrics provide insight into central tendencies, variability, and skewness across countries and years.

| Variable                                                                                                | Mean     | Median   | SD       | Min    | Max      | Skewness |
|---------------------------------------------------------------------------------------------------------|----------|----------|----------|--------|----------|----------|
| <b>Life expectancy at birth, total (years)</b>                                                          | 73.81    | 77.53    | 10.59    | 47.94  | 84.06    | -1.22    |
| <b>GDP per capita (constant 2015 US\$)</b>                                                              | 27782.74 | 11729.41 | 27377.73 | 528.49 | 90605.02 | 0.73     |
| <b>Fertility rate, total (births per woman)</b>                                                         | 2.52     | 1.78     | 1.74     | 1.00   | 7.25     | 1.75     |
| <b>Mortality rate, infant (per 1,000 live births)</b>                                                   | 19.29    | 4.90     | 25.62    | 2.00   | 102.00   | 1.73     |
| <b>Current health expenditure per capita (current US\$)</b>                                             | 2607.16  | 910.44   | 2929.78  | 10.07  | 11262.23 | 1.01     |
| <b>School enrollment, secondary (% gross)</b>                                                           | 86.91    | 94.60    | 30.86    | 10.96  | 151.76   | -0.88    |
| <b>School enrollment, tertiary (% gross)</b>                                                            | 50.94    | 53.67    | 33.59    | 0.79   | 119.68   | 0.12     |
| <b>Carbon dioxide (CO<sub>2</sub>) emissions excluding LULUCF per capita (t CO<sub>2</sub>e/capita)</b> | 5.67     | 5.02     | 5.07     | 0.03   | 19.24    | 1.22     |
| <b>People using at least basic sanitation services (% of population)</b>                                | 77.37    | 95.84    | 31.53    | 9.69   | 100.00   | -1.20    |

Figure 2. Descriptive Statistics

### **3.4. Data Limitations**

While the World Bank Open Data provides consistent and internationally comparable statistics, some limitations exist. First, certain indicators may have missing or irregular data for specific years or countries, which could slightly affect trend accuracy. Second, reporting methods and definitions can vary across countries, particularly for education enrollment rates and CO<sub>2</sub> emissions, potentially introducing inconsistencies. Third, the dataset focuses on national averages, which may mask regional disparities within countries. Finally, secondary data analysis limits the ability to explore causal relationships, as observed correlations do not necessarily imply causation. Despite these constraints, the data remain sufficient for identifying broad trends and comparative patterns across countries.

## **5. App Description**

The Shiny application provides an interactive environment for exploring socioeconomic, health, educational, and environmental trends across countries from 2000 to 2023. You can select one or multiple countries and define a custom year range, and all visual outputs update instantly based on these choices.

The app incorporates ten dynamic, plotly-driven visualizations with interactive tooltips, zooming, and animation. It includes time-series plots such as life expectancy over time, two animated bubble charts showing how GDP relates to life expectancy and secondary education enrollment, scatterplots exploring relationships between education levels, sanitation access, fertility rates, infant mortality, CO<sub>2</sub> emissions, and economic performance. It also applies a logarithmic scale to GDP to improve comparisons between countries with large economic differences.

Finally, the app also includes a correlation heatmap that summarizes the strength and direction of relationships between all key indicators. We ensured consistent numeric formatting that allows accurate plotting and correlation analysis. The interface is combining a simple sidebar for user inputs with a tabbed dashboard for easy navigation across visualizations. In essence, this application offers a comprehensive and interactive tool for examining global development patterns and identifying connections between economic, environmental, and social indicators.

## Socioeconomic and Environmental Trends (2000–2023): A Panel Data Analysis of Selected Countries

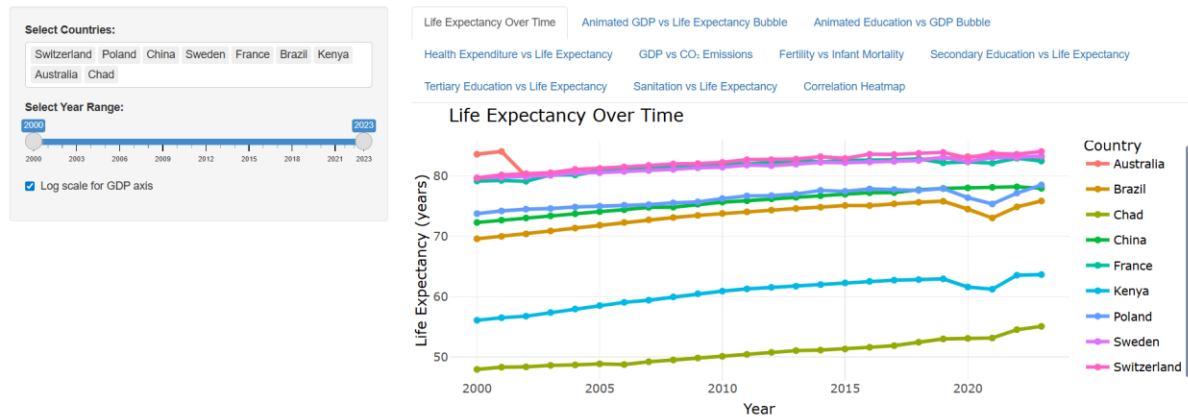


Figure 3. Screenshot of Shiny application and its UI made in RStudio

All figures presented below were generated using our custom-built Shiny application developed in RStudio, ensuring that each visualization is produced directly from our data processing and analytical framework. Based on the insights obtained from this Shiny application, we now proceed to the analysis section.

## 6. Analysis

In this section, we present a comparative analysis of the relationships among the selected variables, aiming to identify both direct and indirect correlations. Particular attention is given to several notable patterns, such as the consistently poor performance of Chad and Kenya across multiple indicators and the markedly superior performance of Switzerland, which aligns with existing expectations.

Our analytical approach incorporates both fundamental statistical techniques and more specialized methods to ensure a comprehensive examination of the dataset. Although every effort was made to include the most relevant and informative analyses, some gaps may remain due to limitations in the source data. The dataset was compiled from multiple sources and subsequently consolidated to enable the most complete and accurate assessment possible.

### 6.1. Life Expectancy Over Time

The life expectancy over time graph illustrates changes in average expected lifespan from 2000 to 2023. As shown, Chad and Kenya initially exhibited notably low life expectancy levels at approximately 50 and 60 years, respectively. However, over the 23-year period, both countries experienced an increase of nearly ten years. This upward shift is broadly consistent with documented international development efforts, including initiatives by humanitarian and non-governmental organizations aimed at improving access to essential resources such as clean water, food security, and sustainable water infrastructure.



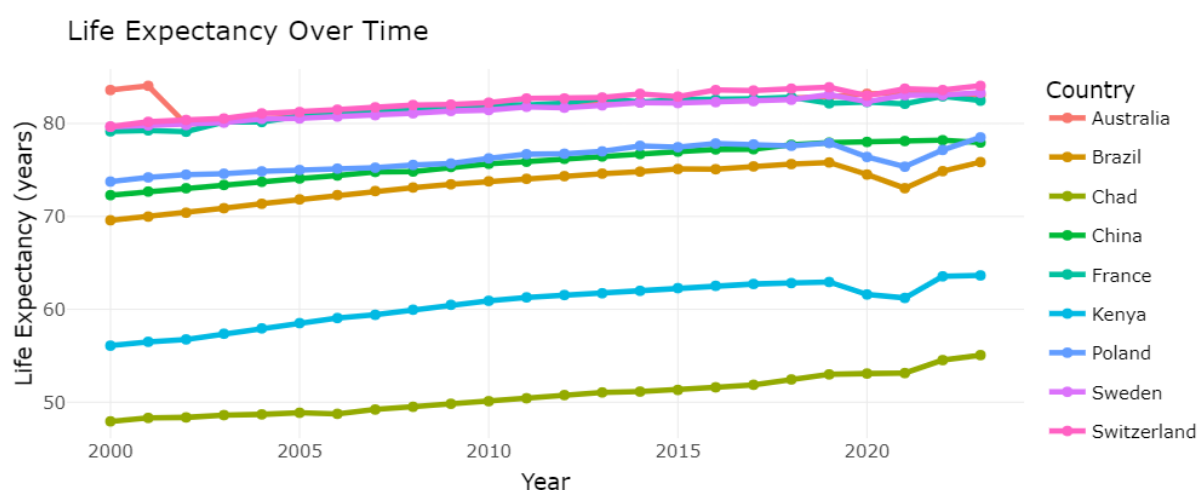


Figure 4. Life Expectancy over time

In contrast, life expectancy in Australia, Switzerland, and France remained relatively stable, with slight upward trends. This pattern aligns with expectations for highly developed countries, where strong healthcare systems, stable socio-economic conditions, and sustained public health policies contribute to already high life expectancy rates.

Overall, the data reveal a clear and positive longitudinal trend: life expectancy has increased across all observed countries. This constitutes one of the most encouraging findings of our analysis.

## 6.2. Health Expenditure and Life Expectancy

The scatter plot comparing health expenditure per capita with life expectancy reveals a clear positive association between the two variables. Countries such as Chad and Kenya, which allocate minimal financial resources to healthcare, exhibit the lowest life expectancy and figures an outcome that aligns with expectations given their limited investment in medical infrastructure and public health services.

Conversely, Australia, France, and Sweden demonstrate a strong balance between health expenditure and life expectancy. These nations typically promote preventive health measures, maintain robust healthcare systems, and prioritize the well-being of their populations, all of which contribute to their comparatively high life expectancy outcomes.

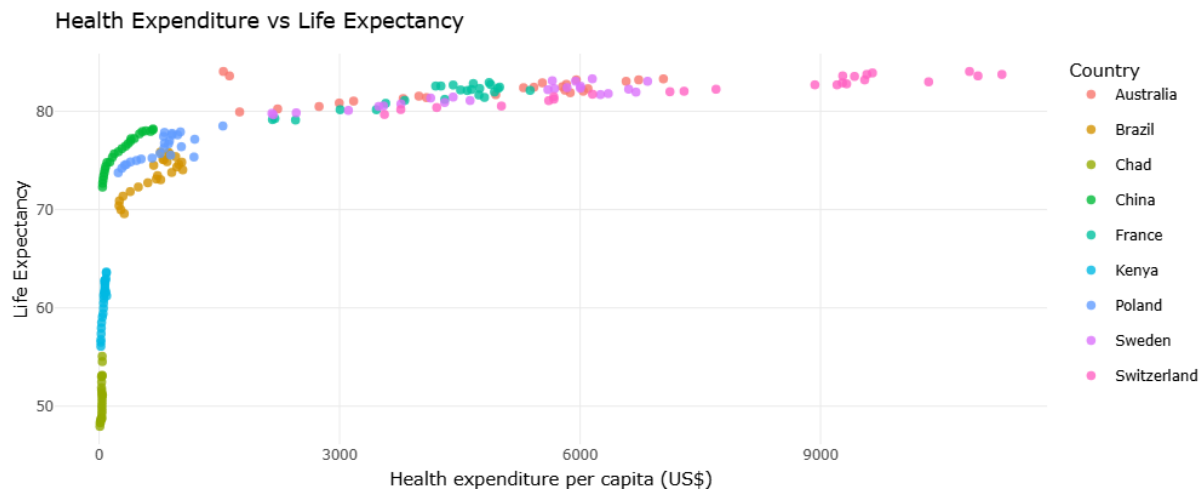


Figure 5. Health Expenditure per capita to Life Expectancy

Switzerland stands out as the most prominent outlier in terms of spending. In several years, its per capita health expenditure exceeded 9,000 USD, with a peak surpassing 11,000 USD in 2021 which is by far the highest among the countries examined.

In summary, the analysis supports a consistent trend: higher per capita health expenditure is associated with longer life expectancy. This relationship reinforces the importance of sustained investment in healthcare as a determinant of population health outcomes.

### 6.3. GDP and CO<sub>2</sub> Emissions

The scatter plot comparing GDP per capita with CO<sub>2</sub> emissions per capita illustrates the relationship between economic output and environmental impact across the selected countries. Unlike some of the clearer associations observed in other parts of our analysis, this plot does not reveal a strong or consistent correlation between the two variables. Countries with comparable GDP levels often display markedly different emission profiles, suggesting that economic wealth alone does not determine environmental burden.

Countries such as Poland and China exhibit relatively high CO<sub>2</sub> emissions despite having moderate GDP per capita. This pattern is consistent with economies that rely heavily on fossil fuels and maintain energy-intensive industrial sectors, both of which significantly contribute to elevated carbon outputs.

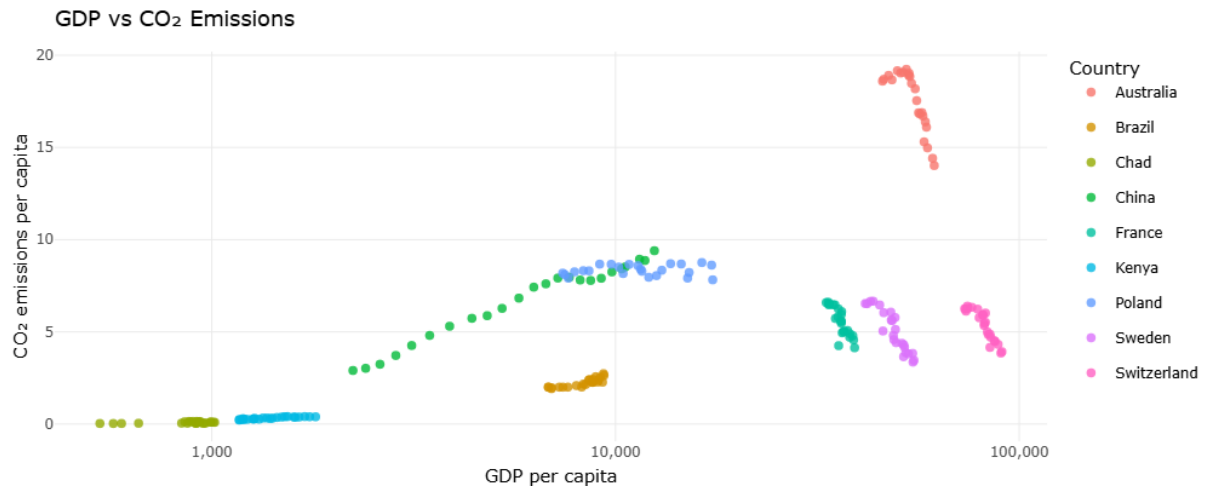


Figure 6. GDP per capita to CO<sub>2</sub> Emissions

In contrast, France, Sweden, and Switzerland demonstrate a more favourable balance between GDPs per capita and CO<sub>2</sub> emissions. These high-income countries maintain comparatively low emissions per capita, which may be attributed to cleaner energy mixes, advanced technological efficiency, and comprehensive environmental regulatory frameworks.

Broadly speaking, the analysis indicates that GDP per capita has limited explanatory power for predicting CO<sub>2</sub> emissions per capita. Instead, emission levels appear to be more strongly influenced by structural factors such as energy dependency, industrial composition, and national environmental policies.

## 6.4. Fertility and Infant Mortality

The scatter plot comparing fertility rates with infant mortality rates shows a clear positive relationship between the two variables. Developing countries such as Chad and Kenya exhibit both high fertility and high infant mortality, largely due to limited healthcare access, poor nutrition, and inadequate medical infrastructure.

Sanitation further contributes to this pattern. Countries like China and Brazil, which have poorer sanitation conditions relative to developed nations, also display higher infant mortality and moderately higher fertility rates.

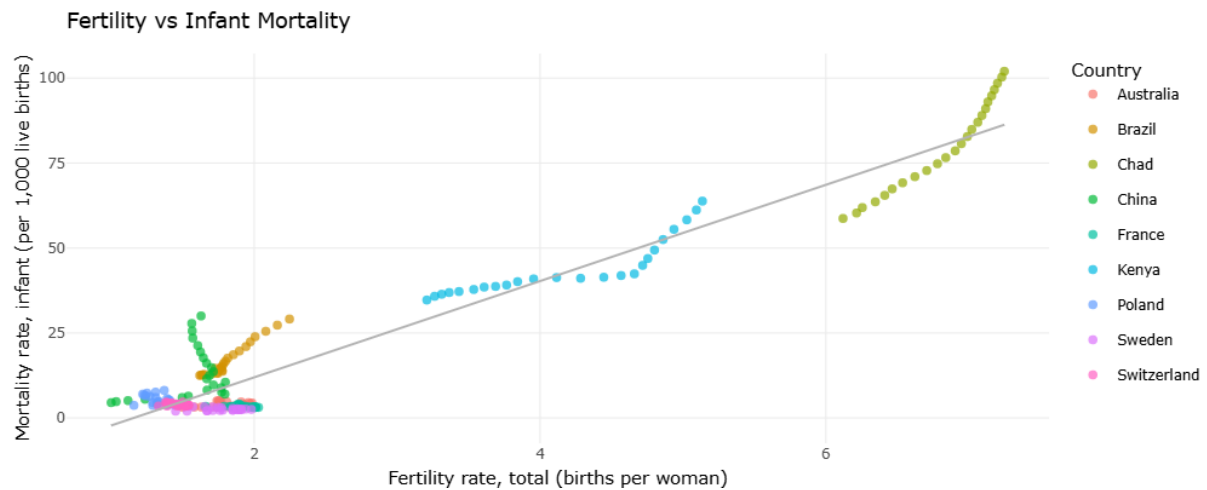


Figure 7. Infant Mortality Rate to Fertility Rate

Additionally, developed countries maintain low fertility and very low infant mortality, supported by strong healthcare systems, effective sanitation, and higher living standards.

In general, the graph indicates that higher fertility tends to coincide with higher infant mortality, especially in countries facing systemic health and sanitation challenges.

## 6.5. Secondary Education and Life Expectancy

The scatter plot comparing secondary education enrollment with life expectancy shows a general positive association, though no strong or surprising trends are observed. Countries with higher access to secondary education tend to exhibit longer life expectancy, reflecting the broader socio-economic benefits of education, including better employment opportunities and access to healthcare and essential resources.

Developed countries typically show higher enrollment rates, supporting this pattern. Anomalies are observed in Sweden, where enrollment exceeded 120% in some years, likely due to

variations in reporting and changes in age cohorts, peaking at 151% in 2000 and fluctuating to 134% by 2023.



Figure 8. Secondary Education School Enrollment to Life Expectancy

Overall, the data suggest that improved access to secondary education is generally linked with increased life expectancy, consistent with established socio-economic theories.

## 6.6. Tertiary Education and Life Expectancy

The scatter plot comparing tertiary education enrollment with life expectancy shows a general positive association, similar to the pattern observed with secondary education. Countries with greater access to higher education tend to have longer life expectancy, reflecting the socio-economic benefits of advanced education, including better career opportunities and improved access to resources that support health and well-being.

Developed and near-developed countries typically exhibit higher tertiary enrollment, which may also contribute to societal advancement through the development of professionals, scientists, and innovators. Anomalies are observed in Australia, where tertiary enrollment exceeded 100% since 2015, peaking at 119% in both 2015 and 2016, likely due to data reporting or cohort effects.

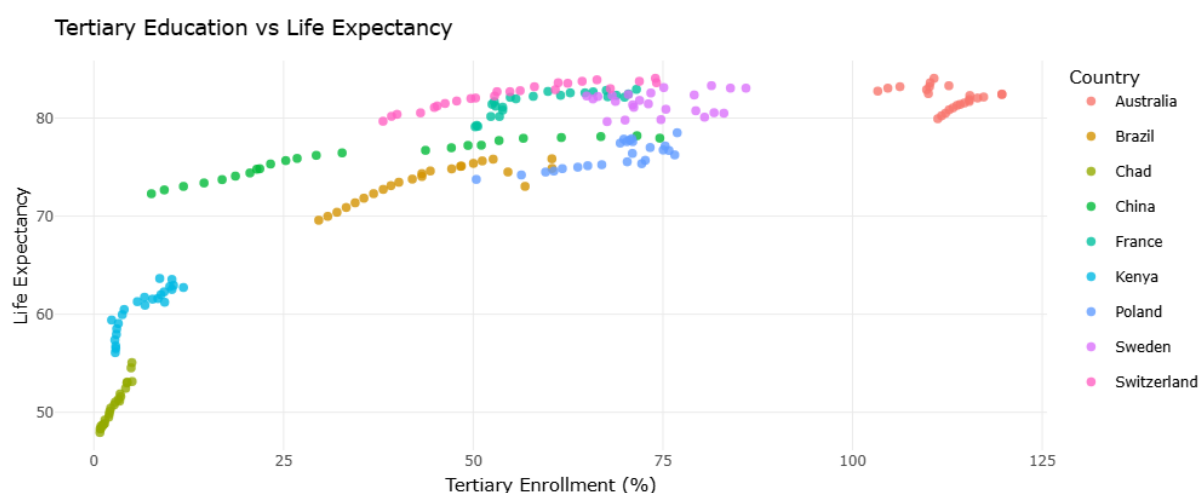


Figure 9. Tertiary Education School Enrollment to Life Expectancy

Summerizing, the analysis reinforces the trend that greater access to higher education is associated with increased life expectancy.

## 6.7. Sanitation and Life Expectancy

The scatter plot comparing sanitation coverage with life expectancy demonstrates a clear and strong positive relationship: countries with better sanitation generally exhibit longer life expectancy. This trend highlights the critical role of public health infrastructure in supporting population longevity.

An exception is observed in France in 2023, where sanitation coverage dropped sharply from 98% in 2022 to 87%. This decline coincided with nationwide sanitation worker strikes over pension reforms, which led to widespread waste accumulation and disruption of services.

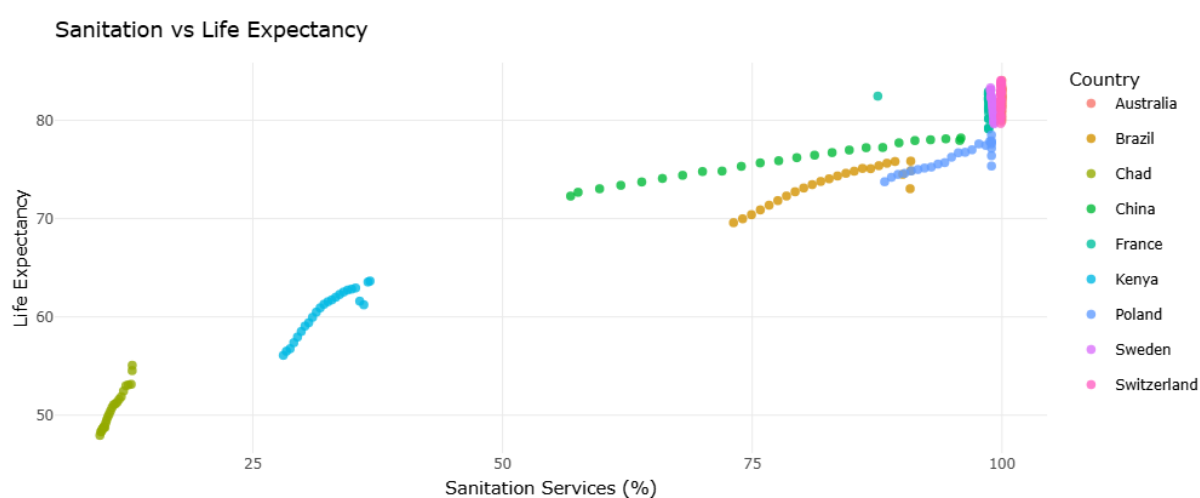


Figure 10. People using at least basic sanitation services to Life Expectancy

Collectively, aside from such temporary anomalies, the data indicate that sanitation is a major determinant of life expectancy. Developing countries such as Chad and Kenya continue to face challenges due to limited funding and insufficient infrastructure, which adversely affect population health.

## 6.8. Correlation Heatmap

Finally, the correlation heatmap illustrates the strength and direction of relationships among the variables in our dataset. Consistent with our earlier observations, the heatmap confirms several key patterns.

For instance, we previously noted the weak influence of GDP on CO<sub>2</sub> emissions, which is clearly reflected in the heatmap. Similarly, there is minimal correlation between health expenditure and CO<sub>2</sub> emissions, which aligns with expectations since much healthcare spending is directed toward routine care rather than carbon-intensive processes.

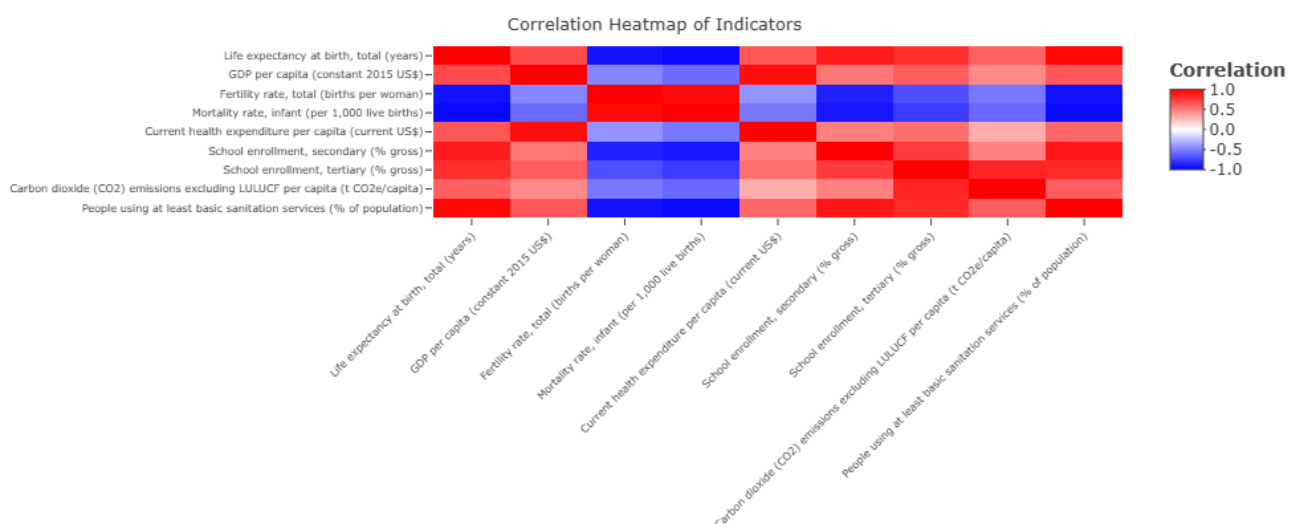


Figure 11. Correlation Heatmap between various indicators

Taken together, the majority of our findings are supported by the correlation analysis, with only a few minor discrepancies. With this comprehensive overview complete, we proceed to present the conclusions drawn from our study and offer recommendations based on the relationships observed among the variables.

## 7. Conclusions and Recommendations

This report examined economic, social, and environmental development across nine countries from 2000 to 2023 through a combination of time-series and cross-country analysis. By using multiple indicators and an interactive Shiny application, the study provides a broad understanding of long-term progress and the remaining disparities between high-, middle-, and low-income countries.

Across all countries, one of the strongest and most consistent patterns is the rise in life expectancy. The improvement is especially notable in Kenya and Chad, where humanitarian support, better access to water, expanded health programs, and gradual infrastructure development have contributed to a significant increase. High-income countries such as Switzerland, Australia, France, and Sweden maintain stable and high life expectancy due to well-funded health systems and a long history of public health investment.

Health expenditure also shows a clear link to population well-being. Countries that invest more in healthcare achieve better health outcomes, whereas nations with limited resources continue to face preventable challenges. Switzerland's very high spending highlights an important point. Once health systems become highly developed, further spending produces smaller gains. This indicates that early and consistent investment is far more impactful than late or excessive spending.

Environmental sustainability presents a more complex picture. The relationship between GDPs per capita and CO<sub>2</sub> emissions is weak and inconsistent. A country's wealth does not determine whether it pollutes heavily or maintains low emissions. Instead, emissions depend on the choice of energy sources, industrial composition, and environmental policy. China and Poland rely more on fossil fuels, while France, Sweden, and Switzerland benefit from cleaner energy structures and effective environmental regulations.

Education, both secondary and tertiary, shows a positive association with life expectancy. Higher levels of education contribute to healthier lifestyles, increased access to information, better employment opportunities, and stronger awareness of hygiene and environmental issues. This supports the idea that education is not only an economic tool but also a foundation for human development.

Sanitation is one of the most influential variables in the entire dataset. Countries with reliable sanitation infrastructure consistently achieve longer life expectancy. The temporary drop in France in 2023 illustrates the essential nature of sanitation services. Even in a highly developed country, disruptions can quickly affect public health and quality of life.

The correlation heatmap confirms that many aspects of development are interconnected. Improvements in education, sanitation, and health tend to reinforce each other, creating stable progress over time. Weakness in one area can slow development in several others. Overall, the



results show that sustainable progress requires coordinated improvements rather than isolated efforts.

Building on these findings, it becomes clear that meaningful progress requires policy strategies that address interconnected areas rather than isolated issues. The relationships observed in the data indicate that improvements in health, education, sanitation, and environmental management reinforce one another, suggesting that development efforts should be coordinated across sectors. With this in mind, several practical recommendations emerge from our analysis.

First, investment in health, education, and sanitation should be approached as an integrated strategy. These sectors are mutually reinforcing: better education enhances health awareness, effective sanitation reduces disease, and improved health outcomes increase school attendance and productivity. Coordinating funding and policy efforts in developing countries would ensure that progress in one dimension strengthens the others and leads to more balanced long-term development.

Second, countries should place greater emphasis on environmentally sustainable economic growth. Since the findings show that GDP per capita alone does not reduce CO<sub>2</sub> emissions, deliberate investment in renewable energy, technological innovation, and cleaner industrial processes is necessary. These efforts would allow countries to maintain economic progress while aligning with global climate objectives and minimizing environmental harm.

Third, expanding access to education should be recognized not only as an economic priority but also as a long-term public health intervention. Higher secondary and tertiary enrollment levels are strongly associated with longer life expectancy, reflecting the role of education in improving health literacy, income opportunities, and informed decision-making regarding hygiene, family planning, and environmental stewardship. Strengthening educational access would therefore contribute to multiple dimensions of development simultaneously.

Finally, the importance of reliable and resilient sanitation infrastructure cannot be overstated. The analysis shows that sanitation coverage is one of the strongest predictors of life expectancy, and disruptions—such as those observed in France during the 2023 sanitation strike—can have immediate and widespread consequences. Countries should invest in robust waste management systems, clean water infrastructure, and contingency plans that ensure continuity of essential services. This applies both to developing countries building foundational systems and to developed nations maintaining and modernizing existing infrastructure.

Together, these recommendations underscore the need for coordinated, multi-dimensional approaches to sustainable development. By addressing health, education, sanitation, and environmental challenges in an integrated manner, countries can strengthen resilience, reduce inequalities, and support the well-being of their populations over the long term.

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